

Project Flow & Entity Dependency Map

Next.js 15 App Router · REST backend · Multi-company · Generated for architecture & QA reference

Sales

Procurement

Inventory

Production

Dispatch

Export

Finance

1. What this system is

The application is the frontend for a garment export house that runs two legal entities (companies). It is a thin Next.js client over a separate REST backend (default base `http://localhost:5001/api`). There is no client-side database or global store: every entity is a backend resource, and each React form component defines the effective schema for its entity (its field object lists create/edit fields, and its dropdowns reveal foreign-key relationships).

Core plumbing

- **API layer** — `lib/api-client.js`. A single `fetchApi()` wraps `fetch`, injects `Authorization: Bearer <token>` from `localStorage.auth_token`, and redirects to `/login` on 401. Exposes `apiClient.get/post/put/patch/delete/download`.
- **Auth & permissions** — `hooks/useAuth.js` loads `GET /auth/me`, flattens role permissions, and exposes `can()` / `canAny()`. The **Super Admin** role bypasses all checks.
- **Multi-company** — almost every entity carries a `company_id`. Masters may be shared (blank company); transactional documents inherit `company_id` from their parent. The UI blocks cross-company mixing.
- **Navigation** — `components/layout/Sidebar.js` is the authoritative module ordering and permission list.

2. Module order (as enforced by the sidebar)

Masters → Sales → Planning → Procurement → Inventory → Production
→ Barcode → Dispatch → Export → Finance → Reports → Administration

3. Entity dependency graph

Arrows read as "depends on / references". A child cannot be created before the parents it points to exist.

Company —(scopes via `company_id`)→ every master & transaction
User —(has roles)→ Role —(has permissions)→ Permission

Buyer → Category[], Agent, Currency, Incoterm, Port, Payment Term
Product → Category, Material Type, UOM, Price Band, GST Rate
Supplier → Supplier Type, Category[], Agent
Jobber → Category[], Product[], Buyer[], Agent
Brand → Company

Inquiry → Buyer, Category, Format, Agent, Currency; items ► Product, Supplier, FOB
└─(convert)→ Order Confirmation (source_inquiry_id)
Order Confirmation → Buyer, Brand, Category, Format, Agent, Currency; items ► Product, Supplier
Purchase Order → Order Confirmation (inherits Company + Buyer) + Supplier; items ► Product
GRN / Inward Entry → Purchase Order (inherits Company); items ► PO line ⇒ creates LOTS
Lot → GRN or Processing (anchor for QC, Stock, Barcode, Dispatch)
Quality Inspection → Lot (⇒ GRN, PO, Supplier)
Supplier Return → Quality Inspection (rejected qty)
Debit Note → Quality Inspection (+ optional Supplier Return); amount uses PO price
Material Issue → Company, Location, Users; items ► Lot (reduces stock)
Processing → issued material ⇒ finished-production LOTS

Barcode → Lot
Dispatch (STOCK) → Order Confirmation + Location; items ► (OC item, finished Lot)
Dispatch (DIRECT) → Purchase Order; items ► PO line
Packing / Export Doc → Buyer, Currency, dispatched goods
Proforma Invoice → Order Confirmation
Invoice → Order Confirmation / Dispatch; items ► dispatch_item_id (price from OC)
Agent Commission → Agent / Buyer / Order Confirmation
Brand Projection → Brand; items ► Product
Material Requirement ► Brand Projection
Material Plan → Material Requirement[]

UI-enforced rules to remember: company inheritance flows down the chain; supplier / buyer / brand dropdowns are filtered by the document's company; the Order Confirmation form reuses the Inquiry create-endpoint for its dropdowns (so it needs the `inquiry.create` permission).

4. The Order Confirmation is the hub

The **Order Confirmation (OC / "Contract")** is the pivot of the whole system. Purchase Orders, Dispatches, Proforma Invoices and Invoices all reference it, directly or indirectly. Confirming an OC unlocks procurement and, later, dispatch and billing against it.

5. End-to-end workflow (order of operations)

Setup

- **1. Administration** — create **Company**, then **Users, Roles, Permissions** (Super Admin bypasses).

Reference data (prerequisites for everything)

- **2. Masters** — Categories → Order Formats → UOM → Material Types → Products → Brands → Buyers → Suppliers → Jobbers → Agents → FOB Values → Markups.

Transaction chain

- **3. Sales — Inquiry** (optional) → convert to **Order Confirmation** → mark **Confirmed**.
- **4. Planning** (optional) — **Brand Projection** → **Material Requirement** → **Material Plan**.
- **5. Procurement — Purchase Order** (against confirmed OC + Supplier) → **GRN / Inward Entry** (creates **Lots**) → **Quality Control** (accept / reject) → optional **Supplier Return + Debit Note**.
- **6. Inventory / Production** — stock lands at a **Location**; **Material Issue** issues QC-accepted lots; **Processing** turns them into finished-production lots.
- **7. Barcode** — generate barcodes for received / finished lots (used when scanning during issue and dispatch).
- **8. Dispatch — Stock Dispatch** of finished lots allocated to the OC (reduces stock), or **Direct Supplier Dispatch** of PO lines.
- **9. Export — Packing List** → **Export Document** (generate shipping PDFs).
- **10. Finance — Proforma Invoice** → **Invoice** (bills posted dispatch lines) → **Buyer Receipts**; on the buy side **Purchase Bills** → **Supplier Payments**, plus **Agent Commission**.
- **11. Reports / Outstanding** — reporting across the whole chain.

6. Stock & money side-effects

ACTION	EFFECT ON STOCK	EFFECT ON MONEY
Post GRN / Inward Entry	Creates lots, increases stock at location	—
Complete QC inspection	Splits accepted / rejected quantity	Enables debit note on rejects
Post Material Issue	Reduces stock (issued to production)	—
Post Processing	Creates finished-production lots	—
Post Stock Dispatch	Reduces finished stock	Enables invoicing of dispatched lines
Direct Supplier Dispatch	No stock movement (mill ships direct)	Enables invoicing
Create Invoice	—	Bills dispatched lines; outstanding tracked
Debit Note	—	qty × PO unit price charged back to supplier

Reference files: `components/layout/Sidebar.js`, `components/sales/order-confirmations/OcForm.js`, `components/procurement/purchase-orders/PurchaseOrderForm.js`, `components/dispatch/DispatchForm.js`, `lib/api-client.js`.